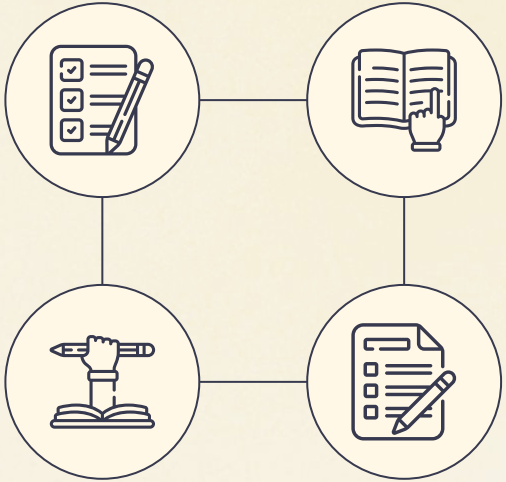
Linguistics in NLP: Evaluating LLM Performance on Core Linguistics Tasks Across Languages

Project Outline

Baseline LLM performance for tasks across models & difficulty



Improve LLM performance on difficult tasks: prompt engineering, LRLs

Design a novel linguistic task to prove LLM generality

Develop conclusions and future directions for research

01

Project Motivation & Statement

Imagine you're a linguistics professor...

You want to identify what a speaker is taking for granted across 10,000 sentences. Can an LLM reliably do that in English? In Mandarin?

How can you do that? You can go to Dr. Choi to ask him to write a pipeline for you, requiring heavy expenditure of resources for a single-use, targetted tool.

LLMs are general-purpose annotation engines: a linguist can describe a task in plain English and an LLM applies that schema without needing technical knowledge, taking advantage of the AI 'black box'.

02

Task-Based Analysis

PROJECT: Baselining

1. **Lemmatization**
   - Proof-of-concept that LLMs can be used to scale linguistics research to avoid manual annotation: they match the SOTA, scoring 96-97% accuracy
2. **POS Tagging & NER**
   - OPUS hits 94% on UD POS tags and 95% F1 on CONLL-2003 NER, even with zero-shot prompting, although performance drops on OOD by about 4-5%
   - Notably, these tasks aren't perfectly defined: these datasets and label sets are considered the gold standard, but sometimes linguistics needs to go beyond that:
     - What if a different language uses different parts of speech?
     - What if you want a label set more complex than PERS, LOC, ORG?

PROJECT: Baselining

3. Presupposition

- Above are Macro-F1 scores. Though English scores lowest and Neutral class collapses in both models, our improvised & fine-tuned methods output probabilistic accuracies of ~80-85% consistently. Performance includes Claude Sonnet 4.6 P0/P1 and ChatGPT 5.4 P0/P2/P4 across 6 languages (DE, EN, FR, HI, RU, VI).

4. Pronoun Resolution

- Multilingual pronoun resolution. Both models fully scored. Prompt engineering has marginal effect in English; larger effect in lower-resource settings, altogether yielding ~85-90% in deterministic evaluation.

5. Grammatically

- Evaluating the CoLA dataset with a gamut of prompt variations gauging statement coherence, binding, agreement, and more. We achieve high 90s in performance accuracy.

PROJECT: Presuppositions

Model	Prompt	DE	EN	FR	HI	RU	VI
Sonnet 4.6	P0	0.82	0.59	0.88	0.86	0.88	0.88
Sonnet 4.6	P1	0.86	0.62	0.88	0.87	0.88	0.88
ChatGPT 5.4	P0	0.86	0.51	0.85	0.75	0.79	0.78
ChatGPT 5.4	P2	0.83	0.57	0.84	0.80	0.78	0.78
ChatGPT 5.4	P4	0.84	0.48	0.92	0.77	0.81	0.77

PROJECT: Pronoun Resolutions

Language	Sonnet 4.6 acc (P0→P4)	GPT 5.4 acc (P0→P4)
English	89% (stable P0–P4)	84–86%
French	88–91%	90–93%
German	86–88%	88–89%
Russian	95–97%	90–93%

PROJECT: Grammaticality Judgements

Prompt	Split	N	Accuracy	MCC	Macro-F1	Coverage	Refusal rate	Invalid rate
direct	in-domain	100	0.88	0.7252	0.8621	1.00	0.00	0.00
anchor	in-domain	100	0.89	0.7453	0.8725	1.00	0.00	0.00
repair	in-domain	100	0.89	0.7453	0.8725	1.00	0.00	0.00
finetuned	in-domain	100	0.89	0.7453	0.8725	1.00	0.00	0.00
checklist	in-domain	100	0.56	-0.0472	0.4762	1.00	0.00	0.00
direct	out-of-domain	100	0.86	0.6638	0.8300	1.00	0.00	0.00
anchor	out-of-domain	100	0.86	0.6616	0.8264	1.00	0.00	0.00
repair	out-of-domain	100	0.86	0.6638	0.8300	1.00	0.00	0.00
finetuned	out-of-domain	100	0.85	0.6357	0.8075	1.00	0.00	0.00
checklist	out-of-domain	100	0.90	0.7615	0.8760	1.00	0.00	0.00

03

Novel Task Design

NARNIA: Task Definition

Agency Labels

Label	Definition
ACTIVE_SPEAKER	Character is speaking or attributed to dialogue
ACTIVE_PERFORMER	Character performs a physical action
ACTIVE_THOUGHT	Character thinks, feels, or has an internal state
ADDRESSED	Character is spoken to or directly addressed
MENTIONED_ONLY	Mentioned by narrator with no agency
MISCELLANEOUS	Group/species label or role that doesn't fit above

Canonical Characters

Canonical Name	Surface Forms
Lucy Pevensie	Lucy · Lu · Daughter of Eve (→ Lucy)
Peter Pevensie	Peter · Son of Adam (→ Peter)
Susan Pevensie	Susan · Su
Edmund Pevensie	Edmund · Ed · Son of Adam (→ Edmund)
Mrs. Pevensie	Mother
Mr. Tumnus	Tumnus · Faun · the Faun
Jadis	the White Witch · the Queen · Her Imperial Majesty
Aslan	the Lion
Prof. Digory Kirke	Professor · Professor Kirke
Mrs. Macready	Mrs. Macready
Father Christmas	Father Christmas

NARNIA: Data Transformation

```
1   Once there were four children whose names were Pete
1   This story is about something that happened to them
    during the war because of the air - raids .
2   They were sent to the house of an old Professor who
    miles from the nearest railway station and two mile
3   He had no wife and he lived in a very large house w
    and three servants .
4   ( Their names were Ivy , Margaret and Betty , but t
5   He himself was a very old man with shaggy white hai
    well as on his head , and they liked him almost at
    came out to meet them at the front door he was so o
    youngest ) was a little afraid of him , and Edmund
    laugh and had to keep on pretending he was blowing
6   As soon as they had said good night to the Professo
    the boys came into the girls ' room and they all ta
7   " We 've fallen on our feet and no mistake , " said
8   " This is going to be perfectly splendid .
9   That old chap will let us do anything we like . "
10  " I think he 's an old dear , " said Susan .
11  " Oh , come off it ! " said Edmund , who was tired
    always made him bad - tempered .
12  " Do n't go on talking like that . "
13  " Like what ? " said Susan ; " and anyway , it 's t
14  " Trying to talk like Mother , " said Edmund .
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1   sentence_id,sentence,entities,chapter
1   1,"Once there were four children whose names were Peter ,
    "[{""text"": ""Peter"", ""canonical"": ""Peter Pevensie""
    {""text"": ""Susan"", ""canonical"": ""Susan Pevensie"",
    {""text"": ""Edmund"", ""canonical"": ""Edmund Pevensie""
    {""text"": ""Lucy"", ""canonical"": ""Lucy Pevensie"", ""
2   2,This story is about something that happened to them whe
    during the war because of the air - raids .,[],1
3   3,"They were sent to the house of an old Professor who li
    ten miles from the nearest railway station and two miles
    "[{""text"": ""Professor"", ""canonical"": ""Professor Di
    ""MENTIONED_ONLY""}]",1
4   4,He had no wife and he lived in a very large house with
    and three servants .,"[{""text"": ""Mrs. Macready"", ""ca
    ""label"": ""MENTIONED_ONLY""}]",1
5   5,"( Their names were Ivy , Margaret and Betty , but they
    )","[{""text"": ""Ivy"", ""canonical"": ""Ivy"", ""label""
    ""Margaret"", ""canonical"": ""Margaret"", ""label"": ""M
    ""Betty"", ""canonical"": ""Betty"", ""label"": ""MENTION
6   6,"He himself was a very old man with shaggy white hair ,
    as well as on his head , and they liked him almost at onc
    he came out to meet them at the front door he was so odd
    youngest ) was a little afraid of him , and Edmund ( who
    laugh and had to keep on pretending he was blowing his no
```

NARNIA: NER Baseline Results

Model	Precision	Recall	F1
sonnet	0.7154	0.7686	0.741
opus	0.7899	0.7769	0.7833
gemini	0.561	0.5702	0.5656
chatgpt	0.5263	0.6612	0.5861

Label	sonnet	opus	chatgpt	gemini
ACTIVE_SPEAKER	0.885	0.916	0.936	0.731
ACTIVE_PERFORMER	0.621	0.909	0.741	0.375
ACTIVE_THOUGHT	0.667	0.667	0.625	0.667
ADDRESSED	0.529	0.571	0.851	0.600
MENTIONED_ONLY	0.702	0.704	0.600	0.348
MISCELLANEOUS	0.000	0.000	0.000	0.000

Model	Precision	Recall	F1
gemini_fewshot	0.7518	0.876	0.8092
sonnet_fewshot	0.7698	0.8843	0.8231
opus_fewshot	0.7518	0.876	0.8092
chatgpt_fewshot	0.731	0.876	0.797

Label	sonnet	opus	chatgpt	gemini
ACTIVE_SPEAKER	0.909	0.916	0.936	0.907
ACTIVE_PERFORMER	0.640	0.783	0.741	0.643
ACTIVE_THOUGHT	0.632	0.632	0.667	0.667
ADDRESSED	0.889	0.864	0.864	0.889
MENTIONED_ONLY	0.750	0.667	0.692	0.691
MISCELLANEOUS	0.800	0.667	0.250	0.800

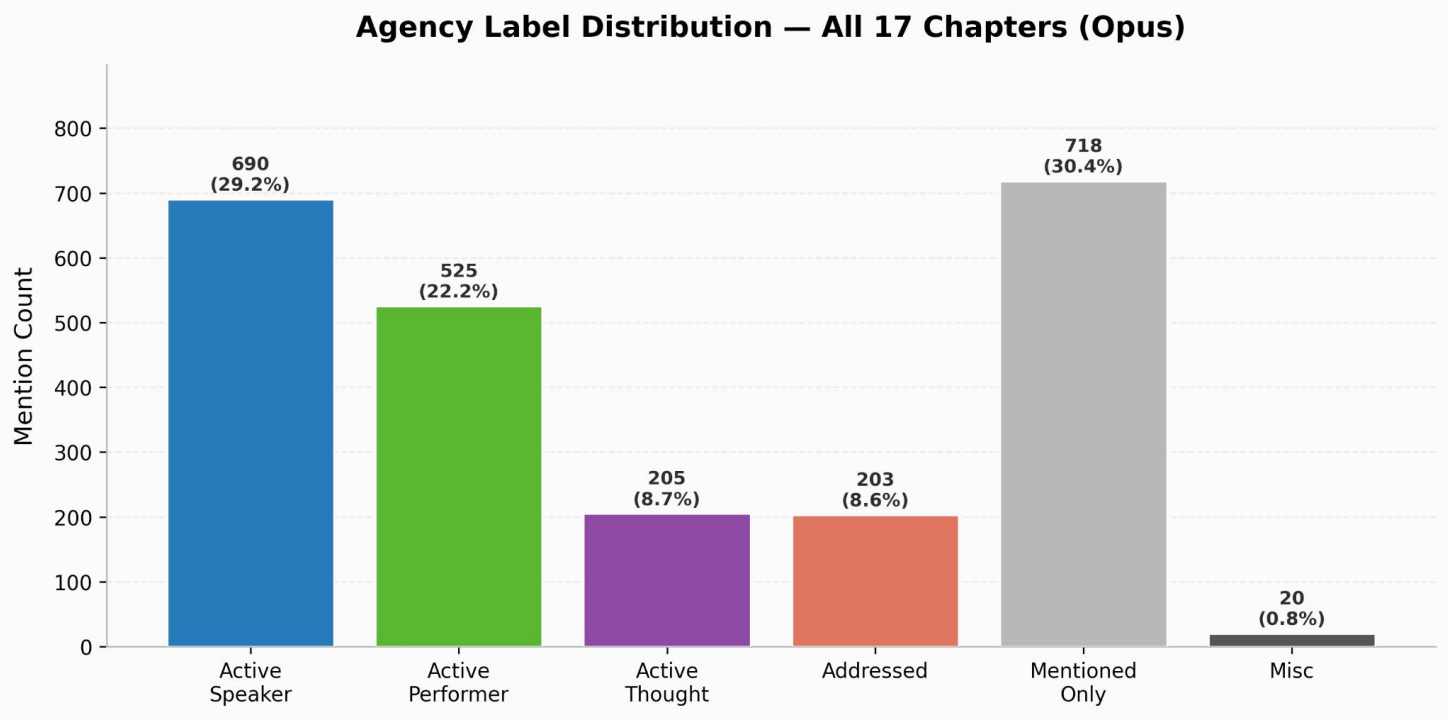
NARNIA: Coreference Baseline Results

Model	Precision	Recall	F1
opus	0.8699	0.8992	0.8843
sonnet	0.7355	0.7479	0.7417
chatgpt	0.2801	0.7815	0.4124

Model	Precision	Recall	F1
sonnet_fewshot	0.7481	0.8487	0.7953
opus_fewshot	0.8	0.8403	0.8197
chatgpt_fewshot	0.4056	0.8487	0.5489

Character	opus	sonnet	chatgpt
Peter Pevensie	1.000	1.000	0.522
Susan Pevensie	0.909	0.909	0.455
Edmund Pevensie	0.889	0.889	0.516
Lucy Pevensie	0.901	0.911	0.462
Professor Digory Kirke	1.000	0.333	0.143
Mrs. Macready	1.000	1.000	0.500
Mr. Tumnus	0.861	0.550	0.342
Jadis	0.800	0.000	0.000
Bacchus	1.000	1.000	1.000
Silenus	0.667	0.667	0.667
Ivy	1.000	1.000	1.000
Margaret	1.000	1.000	1.000
Betty	1.000	1.000	1.000

NARNIA: Corpus Analysis Results



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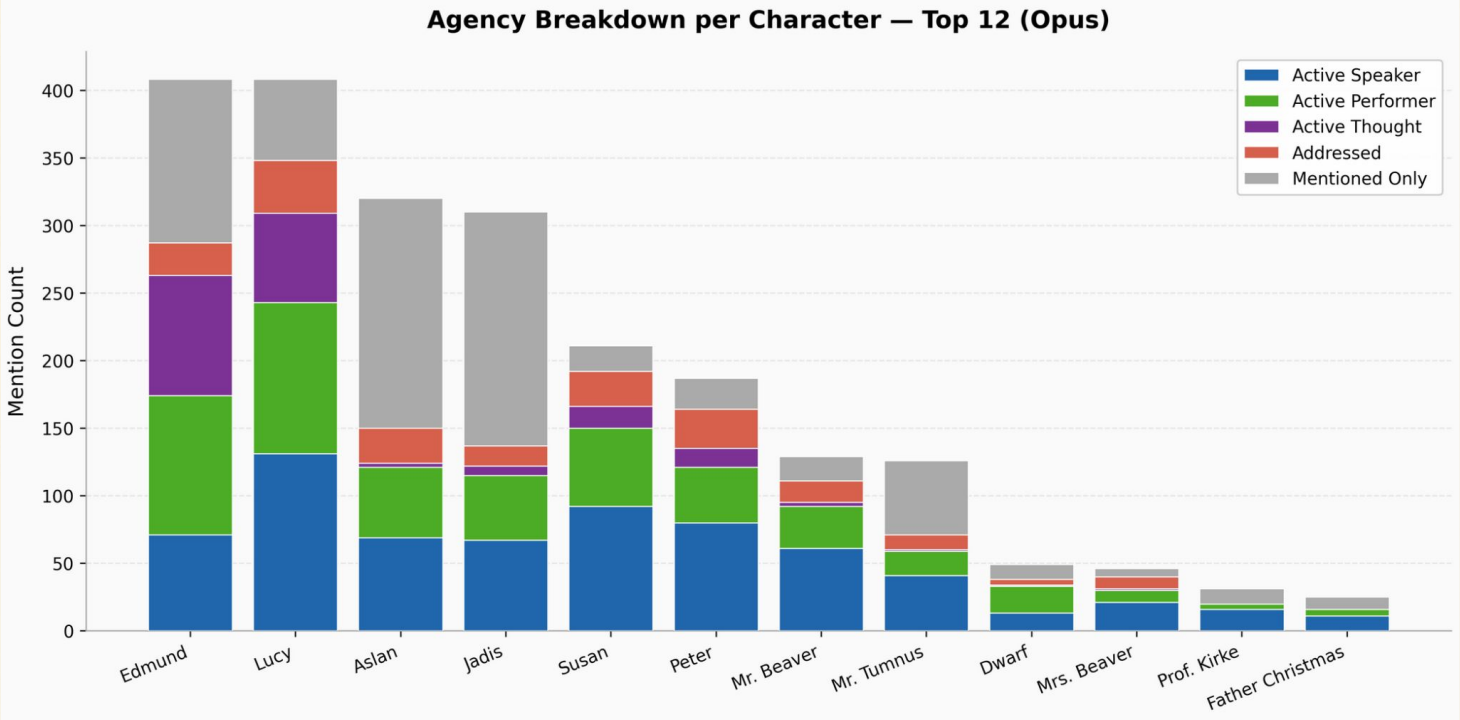

Thank You!